

PocketSmart – Define FastAPI Routes (Backend: FastAPI Integration)

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1. Topic

Defining the FastAPI routes for the PocketSmart backend

Epic: Backend Development

2. Project Overview

PocketSmart is an AI-powered app that helps users track spending, set budgets, and get personalized money-saving suggestions. The FastAPI routes are the entry points that let the user interface send requests to the backend and receive responses.

3. Objective

To define clear, well-organized FastAPI routes that connect user requests to the database, the AI model, and the page templates, with proper validation and error responses.

4. Work Steps

- 1 Plan the list of routes needed for pages, expenses, budgets, insights, and users.
- 2 Create separate router files and group the routes by feature using APIRouter.
- 3 Define the page routes (GET) that render the Jinja2 templates, such as the home page and dashboard.
- 4 Define the expense routes to add, view, update, and delete expenses (POST, GET, PUT, DELETE).
- 5 Define the budget routes to set and view budget limits and savings goals.
- 6 Define the AI insight route that sends the user's spending data to the AI model and returns suggestions.
- 7 Define the user-management routes for registration, login, and profile details.
- 8 Add request validation with Pydantic models and set the correct response models and status codes.
- 9 Add error handling so each route returns a clear message when something goes wrong.
- 10 Register all routers in the main application and test each route using the FastAPI docs page.

5. Main Parts of the Routes

- Router setup and grouping
- Page routes (home, dashboard)

- Expense routes
- Budget and savings goal routes
- AI insight route
- User-management routes
- Validation and error handling